CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

# Code 3: If this, then that

**Goal:** I can make a program react differently when different buttons are pressed.

35 minutes | 7 steps | Computer Science and Programming | Playbook B meeting 28

Ask | Plan | Build | Test | Improve

## You are done when

**Say what happens when A is pressed.**

## You will use

- Cart laptops, charged
- MakeCode for micro:bit in the browser, simulator only, if approved
- Paper if-then cards for the no-device path

## Steps

35 minutes total. Your teacher says when to move on.

### 1 Step 1: Open MakeCode — 4 min

- Get your laptop.
- Open the MakeCode link.

### 2 Step 2: Recap — 1 min

- Tell a partner what a loop does.

### 3 Step 3: Watch a button — 4 min

- Watch on button A pressed change the picture.

### 4 Step 4: Make it react — 14 min

- Show one picture when A is pressed.
- Show another when B is pressed.
- Test both buttons in the simulator.

### 5 Step 5: Trade and test — 4 min

- Test a partner's program.
- Tell them one bug or one win.

### 6 Step 6: Save and shut — 5 min

- Name your project with your initials.
- Close the lid with two hands.
- Put the laptop back and plug it in.

### 7 Step 7: Show your result — 3 min

- Say what happens when A is pressed.

## What to turn in

A program that reacts differently to button A and button B.

## No computer? Another way to do it

Play if-then card games that follow the same rules.

## Finished early? Next challenge

Add a shake event with its own picture.

## Stay safe

Use only the MakeCode link the teacher gives.

## Grade targets

**Grade 5:** Button A and button B show different pictures.
**Grade 6:** Add an if and else condition and test every path.

## Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.